Huancheng Chen

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Education

The University of Texas at Austin, Austin, Texas — 2020 – 2025

Ph.D. in Electrical and Computer Engineering

South China University of Technology, Guangzhou, China — 2015 – 2019

B.E. in Electrical Engineering and Automation

Research Overview

My research interests lie in LLM post-training and efficient LLMs, with a focus on making large language models more capable, reliable, and deployable. Specifically, I am interested in:

- **LLM post-training:** SFT, RL for agentic systems, and training efficiency.
- **Efficient LLMs:** inference acceleration via quantization, pruning, and distillation.
- **Continual learning:** fine-tuning pretrained LLMs and VLMs on downstream tasks.
- **Trustworthy AI:** privacy-preserving machine learning and model safety.

Industrial Experience

Microsoft AI, Mountain View, California — Mar. 2026 – present

Applied Scientist II

- Contributing to training and evaluation of a multimodal relevance ranking model on TB-scale data for Bing.
- Leading development on a Bing-based web agent via SFT / reinforcement learning.

Accenture, Mountain View, California — Jun. 2025 – Mar. 2026

Senior Research Scientist at Advanced AI Center

- Contributed to Accenture's research on memory management for long-horizon LLM agents via MemexRL.
- Built a unified benchmark for agentic and multi-agent reasoning: AgentEvalHub.
- Contributed to the Training-as-a-Service functionality (training, evaluation, and data curation), an end-to-end pipeline that enables clients to customize LLMs with their own data on the AI Refinery platform.

Sony AI, Tokyo, Japan — May. 2024 – Aug. 2024

Research Intern at PPML Team

- Designed MAC, a zero-shot method that enhances spatial semantics in diffusion-based layout-to-image generation.

Sony AI, Austin, Texas — Feb. 2024 – May. 2024

Research Intern at PPML Team

- Proposed DualLoRA, a novel PEFT method for continually fine-tuning pretrained ViTs on downstream tasks.

Toyota InfoTech Lab, Mountain View, California — May. 2022 – Aug. 2022

Research Intern at Infrastructure & Data Platform

- Applied knowledge distillation in FedHKD to mitigate the model drift problem in non-IID distributed learning.

Bell Labs, Murray Hill, New Jersey — Jan. 2022 – May. 2022

Research Intern at Mathematics & Algorithms Research Group

- Developed a U-2-Net-based framework for removing irrelevant background to improve flaw detection.

Publications

($\star$ indicates equal contribution)

[ICLR] D. Kim, S. Cha, **H. Chen**, C. Wang, H. Vikalo, "Quantized Gradient Projection for Memory-Efficient Continual Learning", *International Conference on Learning Representations*, 2026.

[CVPRW] **H. Chen**, J. Li, W. Zhuang, C. Chen, L. Lyu, "Replay-Free Continual Low-Rank Adaptation with Dynamic Memory", *Computer Vision and Pattern Recognition Workshop*, 2026.

[CVPRW] **H. Chen**, J. Li, W. Zhuang, H. Vikalo, L. Lyu, "Training-Free Layout-to-Image Generation with Marginal Attention Constraints", *Computer Vision and Pattern Recognition Workshop*, 2026.

[NeurIPS] **H. Chen**, H. Vikalo, "Heterogeneity-Guided Client Sampling: Towards Fast and Efficient Non-IID Federated Learning", *Advances in Neural Information Processing Systems*, 2024.

[ICML] **H. Chen**, H. Vikalo, "Recovering Labels from Local Updates in Federated Learning", *International Conference on Machine Learning*, 2024.

[CVPR] **H. Chen**, H. Vikalo, "Mixed-Precision Quantization for Federated Learning on Resource-Constrained Heterogeneous Devices", *Computer Vision and Pattern Recognition*, 2024.

[ICLR] **H. Chen**, H. Vikalo, J. Wang, "The Best of Both Worlds: Accurate Global and Personalized Models through Federated Learning with Data-Free Hyper-Knowledge Distillation", *International Conference on Learning Representations*, 2023.

[CVPRW] **H. Chen**, H. Vikalo, "Federated Learning in Non-IID Settings Aided by Differentially Private Synthetic Data", *Computer Vision and Pattern Recognition Workshop*, 2023. **Oral**

[ICCVW] A. Mohamed*, **H. Chen***, "Skeleton-Graph: Long-Term 3D Motion Prediction From 2D Observations Using Deep Spatio-Temporal Graph CNNs", *International Conference on Computer Vision Workshop*, 2021.

Preprints

($\star$ indicates equal contribution)

Z. Wang, **H. Chen**, J. Wang, W. Wei, "Memex(RL): Scaling Long-Horizon LLM Agents via Indexed Experience Memory", *Technical Report*.

S. Cha, **H. Chen**, D. Kim, H. Zhang, K. Chan, G. Veciana, H. Vikalo, "Regularized Calibration with Successive Rounding for Post-Training Quantization", *Under Review*.

H. Chen*, S. Cha*, H. Vikalo, "Task-Agnostic Federated Continual Learning via Replay-Free Gradient Projection", *Under Review*.

Skills

- Platforms: Linux, Azure, AWS, GCP
- Programming: Python == LaTeX== Matlab > Java > C/C++
- Agent & training stacks: Torch, VeRL, Slime, SgLang, vLLM, Ray, Skypilot, Kubernetes

Community Services

- Reviewer for Conference:
  ICML (22,23,24,25,26), NeurIPS (22,23,24,25,26), ICLR (24,25), IJCAI (24,25),
  AAAI (25), CVPR (24,25).
- Reviewer for Journal:
  IEEE Transactions on Mobile Computing